BASINMARK: WATERMARKING DIFFUSION LANGUAGE MODELS VIA KEYED RE-DENOISING CONTRAST

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ABSTRACT

Working draft. This document exists to typeset and check the baseline comparison while the experiments are still running; the prose is deliberately minimal. Every number is measured on one TITAN RTX unless a row says otherwise, and rows that have not been measured are marked pending rather than left blank or filled in from a paper.

1 BASELINE COMPARISON

Table 1 reports watermark detectability against the text quality actually paid for. The axis that matters is not the largest achievable z: an earlier revision of this work tuned an operating point to $z \approx 5$ and only afterwards measured that it cost $4.8\times$ perplexity. Detection is therefore reported as the true positive rate at a controlled false positive rate, and quality as perplexity relative to a non-watermarked control produced under the same decoding regime.

Two properties of the comparison are worth stating before the numbers. *First, each block of the table has its own non-watermarked control.* The three methods require different decoding regimes — dgMARK uses block decoding with top-k sampling, KGW requires strict left-to-right decoding because its green list is seeded by the preceding token, and BasinMark operates post-hoc on drafts sampled at temperature 0.8 over the full vocabulary — so a single shared baseline would charge the cost of a decoding regime to whichever watermark happened to use it. *Second, detection cost is a column, not a footnote.* dgMARK and KGW detect by hashing token identities and need zero model forward passes; BasinMark's detector must run the diffusion model itself. Collapsing the comparison to detectability alone would hide the method's main structural disadvantage.

Reading the table. The strongest baseline, dgMARK with 3-beam lookahead, reaches 0.86 TPR at 1% FPR for $1.23\times$ perplexity. BasinMark's best configuration inside a comparable quality budget reaches 0.10, and its best configuration at any quality reaches 0.50 while paying $2.49\times$. The gap is roughly $8\times$ in detectability at matched quality, which is far too large to attribute to hyperparameter search.

The measured cause is capacity rather than tuning. Across all 36 BasinMark configurations the median carrier position admits exactly *one* token within the fluency budget, and the candidate slate never truncates, so a watermark defined on *which tokens are present* has almost no freedom to spend. This also explains why moving the watermark to generation time made things worse rather than better: the schedule it forces — all non-carrier positions, then all carrier positions — costs $1.98\times$ perplexity on its own, before any watermark is applied.

Method	Setting	PPL ↓	Ratio	TPR at FPR 5%	TPR at FPR 1%	TPR at FPR 0.1%	z	Det. fwd. ↓
Generation-time watermarks (the watermark is applied while the text is produced)								
No watermark	block dec., top-k 3	9.94	1.00×	–	–	–	+0.04	–
dgMARK (Hong & No, 2026)	$k = 1$	15.40	1.55×	0.88	0.74	0.62	+4.35	**0**
dgMARK (Hong & No, 2026)	+3-beam	12.23	1.23×	**0.92**	**0.86**	**0.82**	+5.81	**0**
No watermark	left-to-right	11.61	1.00×	–	–	–	+0.00	–
KGW (Kirchenbauer et al., 2023)	$\delta = 1$	pending	–	–	–	–	–	**0**
KGW (Kirchenbauer et al., 2023)	$\delta = 2$	pending	–	–	–	–	–	**0**
KGW (Kirchenbauer et al., 2023)	$\delta = 3$	pending	–	–	–	–	–	**0**
Post-hoc watermarks (the watermark is applied to text that already exists)								
No watermark	$T = 0.8$, full vocab	26.21	1.00×	–	–	–	−0.03	–
BasinMark	keyed carrier	27.7	1.11×	0.00	0.00	0.00	+0.73	9
BasinMark	entropy gate	41.7	1.59×	0.30	0.20	0.10	+1.49	9
BasinMark	leverage, quality-first	30.5	1.17×	0.10	0.10	0.00	+0.58	9
BasinMark	leverage, detection-first	65.3	2.49×	0.62	0.50	–	+2.94	9
BasinMark	generation-time, two-phase	203.7	7.77×	0.12	0.00	0.00	+1.14	9

Table 1: Watermark detectability against the text quality actually paid for. All rows measured on LLaDA-8B-Instruct, C4 prompts (300-character truncation), 256 generated tokens, GPT-2-large perplexity; $n = 50$ for dgMARK, $n = 30$ for KGW, $n = 10$ for BasinMark. *Ratio* is perplexity relative to the matching non-watermarked arm directly above it. Each block is compared against its own control because the decoding regime differs: dgMARK uses block decoding with top-k 3, KGW requires strict left-to-right decoding (its green list is seeded by the preceding token, which order-agnostic decoding does not provide), and BasinMark operates on drafts sampled at $T = 0.8$ over the full vocabulary. *Det. fwd.* is the number of model forward passes a detection requires. Non-watermarked arms sit at $|z| \leq 0.05$ with unit standard deviation, so the thresholds are not inflated. Pending rows are re-running after a detector fault: scoring repeated bigrams gave a 37% false-positive rate at $\delta = 0$ under forced left-to-right decoding, and each distinct bigram is now scored once.

References

Pyo Min Hong and Albert No. dgMARK: Decoding-Guided Watermarking for Diffusion Language Models. In *International Conference on Machine Learning (ICML)*, 2026. arXiv:2601.22985.

John Kirchenbauer, Jonas Geiping, Yuxin Wen, Jonathan Katz, Ian Miers, and Tom Goldstein. A watermark for large language models. In *International Conference on Machine Learning (ICML)*, 2023.